

Sex-Specific Adolescent Development of the Trochlea and Distal Femoral Pysis

A 3D Shape Modeling Study

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Background: Trochlear dysplasia is a primary risk factor for patellar instability and is associated with high recurrence rates, early arthritis, and poor return-to-sport outcomes. Surgical correction with trochleoplasty can improve tracking but carries risks in skeletally immature patients, including physeal damage and growth disturbance.

Purpose: To inform safer surgical decision-making and refine current treatment strategies with a clearer understanding of how the trochlea and pysis develop during adolescence is needed.

Study Design: Cross-sectional study; Level of evidence 3.

Methods: The authors retrospectively analyzed 261 nondysplastic knee magnetic resonance images from individuals aged 8 to 22 years. The authors trained a deep learning model to segment diaphyseal and epiphyseal femoral structures, and then constructed sex-specific statistical shape models of distal femoral shape and pysis location. Regression analyses indicated a typical progression of femur shape with age.

Results: Between ages 13 and 19 years, both sexes demonstrated proximal migration of the pysis and deepening of the trochlea, with greater lateral trochlear depth than medial trochlear depth. Across this age range, medial and lateral trochlear growth progressed at similar rates. After age 16, males demonstrated a significantly greater size-standardized lateral trochlear depth than females.

Conclusion: These findings enhance understanding of sex-specific trochlear development and physeal positioning while demonstrating the utility of deep learning-based 3-dimensional modeling for characterizing knee growth.

Keywords: patellofemoral instability; statistical shape modeling; trochlear dysplasia; trochleoplasty

Trochlear dysplasia is a significant risk factor for patellar dislocation, present in up to 85% of patients with patellar instability.⁸ Left untreated, patellar dislocation has a high recurrence rate, with up to 49% of patients experiencing recurrent episodes after an instability event.^{4,10,22} These patients face increased arthritic risks, and 44% to 60% fail to return to sport at 6 months after a primary dislocation.⁷

Trochlear dysplasia reduces resistance to lateral patellar translation, elevating the risk of dislocation.⁴ Consistent with the finding that patellar dislocation frequently occurs laterally, previous research by Richmond et al²⁶ and Ayala et al³ shows that typical pediatric trochlear morphology provides lateral resistance through a greater lateral trochlear depth compared to medial trochlear depth, a feature observed as early as 2 years old. Moreover, previous research has identified sex-relevant distinctions in patellar instability events. Patellar dislocation occurs at higher rates in females compared to males,²⁷ and females of all ages experience higher redislocation rates, suggesting underlying morphological differences between the

sexes.¹⁵ Previous research by Pagliazzi et al²⁵ supports this, showing that females typically have a shallower trochlea than males from 1 month to 10 years of age, contributing to a higher risk of dislocation.

Trochleoplasty, a surgical procedure performed to create a more stable trochlear groove for patellar tracking, can reduce instability.¹⁹ While it has been shown to reduce the incidence of patellar dislocation, it is also associated with a 33% increase in reported pain relative to a nonoperative baseline, a 30% chance of developing osteoarthritis, and potential complications, such as physeal damage and growth disturbances in children.^{4,32}

Given the high-risk nature of trochleoplasty, there is growing interest in developing modifications to trochleoplasty or safer alternatives for treating trochlear dysplasia. Achieving a deeper understanding of the relationship of physis to trochlear morphology during development may aid not only in exploring novel approaches but also in improving current surgical decision-making. Such knowledge can inform trochleoplasty, soft tissue procedures, and other surgical developments by identifying age- and sex-specific anatomical landmarks, trochlear configuration, and physeal positions to guide surgical interventions with lower risk of complications.

The prototypical development of the trochlea is understudied because of the limited availability of pediatric cadaveric tissue for physical investigation. An alternative approach is to extract anatomical surfaces from medical images. One promising approach is statistical shape modeling (SSM), a method that represents complex 3-dimensional (3D) anatomical structures with a series of independent shape features. These shape features represent shape characteristics present in the data and can be readily leveraged for statistical analysis.²¹ For example, they can be used to model age-related morphological variations^{12,17} or to examine 3D differences in knee cartilage that may indicate early degenerative changes associated

with osteoarthritis.¹³ In addition to producing shape features for statistical analyses, an SSM is a generative model. It can thus generate synthetic anatomical surfaces to visualize how shape changes as a function of biological factors such as age.^{13,14,17}

This study aimed to develop a 3D statistical shape model of the adolescent femur and use it to measure how trochlear depth and distal femoral physeal position change—and covary—through adolescence.

METHODS

This study was approved by our Institutional Review Board. Given its retrospective nature and use of de-identified imaging data, the requirement for informed consent was waived.

Dataset Description

Pediatric knee magnetic resonance images (MRIs) were identified through a systematic search of our institution's Picture Archiving and Communication System (PACS) for all knee MRIs of patients aged 8 to 23 years acquired between January 2009 and April 2023. Since we aimed to model healthy trochlear development, all those with signs of trochlear abnormality were excluded by screening radiology reports for mention of trochlear dysplasia, patella alta, general patellofemoral joint dysplasia, patellar dislocation, patellar tendon injury, medial patellofemoral ligament injury, and patellar maltracking. Images with missing radiology reports were excluded. After description-based exclusion, MRIs were visually examined, and any patients without visible physes or with hardware indicative of knee surgery were excluded.

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Ethical approval was obtained from the Stanford University Institutional Review Board (IRB) (IRB-65352), and the requirement for informed consent was waived due to the retrospective design.

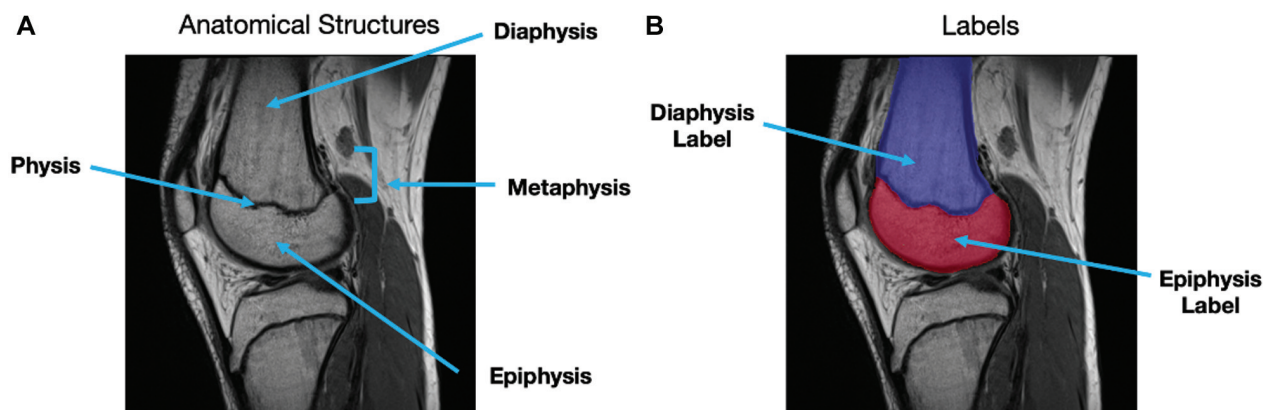


Figure 1. Segmentation Labels. An example MRI slice of the distal femur annotated with (A) anatomical regions: diaphysis, metaphysis, physis, and epiphysis, and (B) corresponding segmentation labels used to delineate regions of the distal femur via segmentation. The diaphysis label contains the diaphysis, metaphysis, and physis. The epiphysis label contains only the epiphysis. MRI, magnetic resonance imaging.

MRI Data and Segmentation

Proton density-weighted MRIs balance tissue contrast and anatomical detail. As scans were collected over a 14-year window from different clinical settings, magnetic field strength and sequence parameters varied across the cohort (as described in Appendix Table A1). However, all sequences maintained sufficient spatial resolution and image quality for consistent segmentation and analysis. For each proton density-weighted MRI, we segmented 2 distinct osseous regions. The first region encompassed the physis along with all proximal structures—including the diaphysis and metaphysis—while the second region comprised the distal epiphysis (Figure 1). The boundary between labels was interpreted as the distal region of the physis. We applied an iterative semiautomated segmentation approach, similar to previous studies and detailed^{14,33} in Appendix 2. Proton density-weighted sagittal MRIs were used for segmentation because this sequence provided the most consistent and highest-quality visualization of the distal femoral physis across our retrospective cohort. Although segmentation was performed on sagittal acquired MRI data, the segmentation model operated on volumetric data, preserving full 3D spatial context.

Statistical Shape Model of Distal Femur Shape and Physis Location

We extracted a mesh of the femur surface from each segmentation using established methods.^{13,14} We then labeled each vertex on the bone surface mesh as epiphysis or diaphysis based on the label of the closest segmentation voxel (Figure 2). This approach enabled the creation of a 3D bone surface that incorporates the distal physis. Notably, this combination of both the distal femur's shape and the distal physis' position uniquely enables modeling of the developmental relationship between overall shape and physis location during adolescence.

To learn the shape and physis location features of our dataset, an SSM was fit to the osseous surfaces using previously established methods^{13,14} (detailed in Appendix 3.1). The purpose of an SSM is to capture variation in distal femoral curvature and physis position independent of overall size. Briefly, fitting the SSM includes 3 steps. The first step is rigid-plus-scaling (similarity) registration. This registration aligns all femurs into a single orientation, removing differences in scale, rotation, and position among them. The second step employs nonrigid registration to obtain matching points between all femurs. Specifically, nonrigid registration morphs a reference bone surface mesh onto each other surface mesh in the dataset, establishing corresponding points within the same anatomical regions across subjects. The third and final step employs principal component analysis to extract shape features. Principal component analysis is a dimensionality reduction technique that converts the thousands of vertices on each subject's bone surface into a set of principal components, which we refer to as shape features. These shape features represent the modes of greatest variation among femurs in the dataset, after removing size. Because of sex differences in aging and trochlear development, it was unclear whether there should be sex-specific SSMs. We built and compared female-only, male-only, and mixed-sex SSMs and determined that the mixed-sex model outperformed the sex-specific models for anatomical surface reconstruction (detailed in Appendix 3.2). Therefore, all analyses were performed using a single SSM trained on all available data from both sexes.

Generating Age-Specific Prototypical Digital Femurs via Linear Regression

To model prototypical trochlear development, we trained a linear regression model to predict age from SSM shape features as input. Because of sex-based differences in anatomical size^{11,24} and development,²⁰ it was unclear

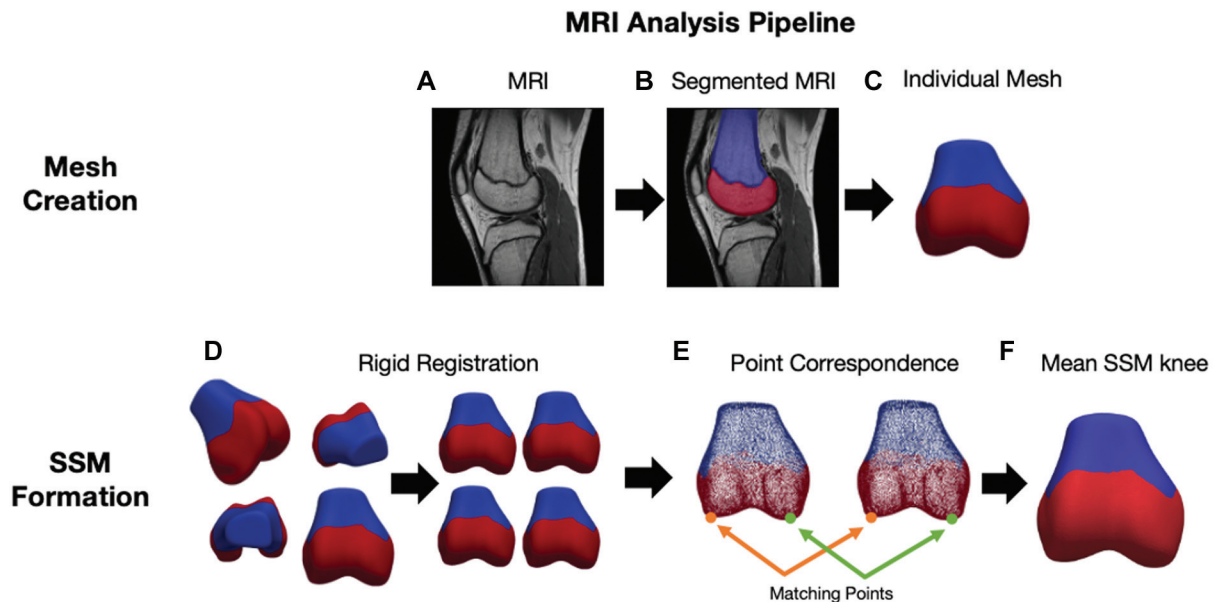


Figure 2. MRI analysis pipeline. (A) Representative sagittal slice from the volumetric MRI of the distal femur. (B) Segmented MRI. After training iterations, a machine learning model (nnUNet, cascaded 3D)¹⁶ segments the diaphysis (blue) and epiphysis (red) from volumetric MRI data. (C) 3D meshes reconstructed from segmentations across individuals. Slice segmentations from each volume are combined and smoothed, and surfaces are extracted using the marching cubes algorithm. SSM Formation: (D) Similarity registration, in which all meshes are aligned into a shared coordinate system and global scale differences are removed. (E) Nonrigid registration, in which each mesh is assigned an identical number of vertices with 1-to-1 point correspondence across all subjects, establishes homologous anatomical locations. (F) The mean SSM knee was generated by averaging shape features across the dataset. The mean knee is shown here for illustration, although the SSM represents the full spectrum of anatomical variation in the dataset. MRI, magnetic resonance imaging; SSM, statistical shape modeling.

whether females and males should share the same linear regression model. We therefore developed sex-specific and mixed-sex regression models and compared their abilities to predict age using shape features as the independent variables. Sex-specific regression models for each sex improved age prediction from shape features, as evidenced by higher R^2 values (detailed in Appendix 4.1). It is important to note that a single mixed-sex SSM was used to capture all shape variations. Sex-specific regression models were then applied separately to the shared SSM to estimate the rate and direction of change in shape features with age, generating 2 sets of sex-specific prototypical meshes. All presented analyses, therefore, use a shared SSM with sex-specific regression models to represent prototypical aging. To validate both the SSM and regression model, held-out subjects were used to evaluate reconstruction accuracy and age prediction performance, respectively. Full details are provided in Appendix 3.2 and 4.1.

The fitted linear regression coefficients can be interpreted as a line, or vector, in shape space that defines average aging. Leveraging the generative nature of the SSM, we systematically varied shape by moving along the fitted aging line and generating 3D surfaces of the average patient as a function of age. Using this approach, we generated prototypical shape representations for males and females aged 13 to 19 years, the 10th and 90th percentiles of ages in our dataset. Our processing pipeline removed

overall size; thus, our model only reflects changes in relative shape as a function of age and does not account for uniform increases in size during aging or between sexes.

Measures of Bone Morphology and Physis Location

To assess the change in trochlear depth and physis positioning with aging, lateral and medial trochlear depths and distances from the distal femur to the anterior, medial, and lateral physis were measured in ParaView.¹ Because all meshes were similarity-registered into a shared coordinate frame as part of the SSM fitting process before analysis, measurements were performed automatically using a consistent plane across all subjects. They thus did not depend on visualization or coordinate-system-based differences (Figure 3). Although the SSM captures full 3D femoral morphology, trochlear depth and physeal distance measurements were intentionally extracted as scalar measures along these fixed anatomical axes, allowing us to summarize developmental changes in 3D shape in an interpretable form.

The lateral and medial trochlear depths were defined as the distances in the anterior direction from the deepest point of the trochlear groove to the highest point of the corresponding medial or lateral crest. The distal femur to physis measurements were defined as the distances in

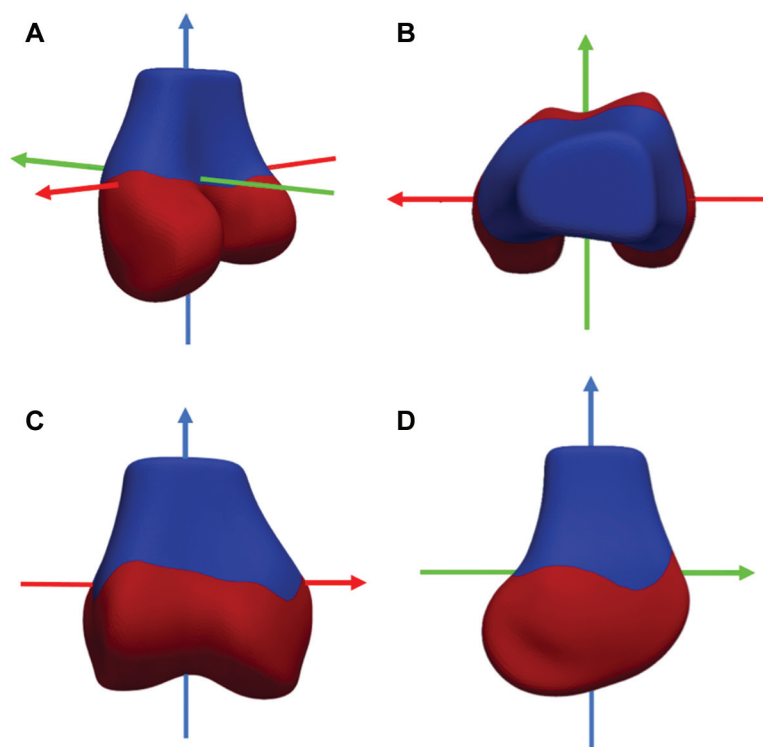


Figure 3. Orientation of the mean femoral mesh within the standardized coordinate system after similarity registration. (A) Oblique view showing all 3 anatomical axes: green = AP; red = ML; blue = SI. This shared coordinate frame is consistent across all registered meshes, enabling standardized, orientation-independent measurements. (B) Distal view looking along the superior-inferior (blue) axis. (C) Anterior view looking along the anterior-posterior (green) axis. (D) Lateral view looking along the medial-lateral (red) axis. AP, anterior-posterior; ML, medial-lateral; SI, superior-inferior.

the superior direction from the most distal point of the femur to the most distal point on the diaphysis label along the anterior, medial, and lateral regions of the trochlea. The most distal point was defined algorithmically as the surface point with the most extreme coordinate value along the superior-inferior axis of this standardized coordinate system. Because every mesh had 1-to-1 point matching, this point was automatically identified in all subjects without visual selection. Examples of these measurements are shown in Figure 4.

Data and Statistical Analysis

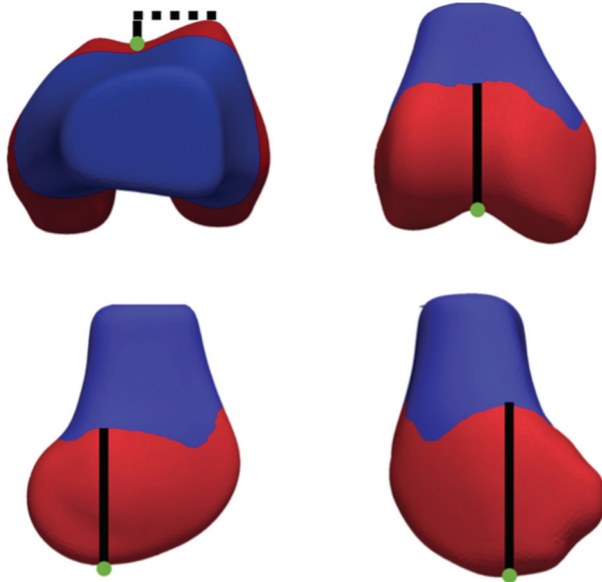
Our primary objective was to quantify age-related differences in trochlear morphology and physis position, and to assess their interrelationship, in an anatomically normal population. Using the fitted aging linear regression models derived from the SSM, we visualized how shape evolved with age. For each sex, we applied the mean model coefficients to generate age-representative meshes of ages 13 through 19 years. To construct 95% CIs, we used the standard error of each regression coefficient to compute upper and lower bounds at the 2.5th and 97.5th percentiles. These fitted regression models were used to generate the mean, 2.5th percentile, and 97.5th percentile trajectories of bone shape as a function of age. Each measurement—lateral

and medial trochlear depths, as well as lateral, medial, and anterior physis distances—was extracted from each generated mesh. Statistical differences between the sexes were assessed by examining whether the 95% confidence bands overlapped. Statistical significance of age-related changes was assessed by fitting linear regression models of both trochlear and physis measurement against age; a regression slope with $P < .05$ was considered evidence of a significant age effect.

To understand the relationship between trochlear development and physeal position, we performed Pearson correlation analyses. Specifically, we correlated the lateral and medial trochlear depths with each of the anterior, medial, and lateral physis distance measures. Correlations were computed separately for males and females. Correlation strength was interpreted using standard thresholds: negligible ($r < 0.10$), weak (0.10-0.39), moderate (0.40-0.69), strong (0.70-0.89), and very strong (≥ 0.90) as described by Schober et al.²⁸

RESULTS

Among 569 MRIs, 261 met the inclusion criteria. The age range of included MRIs was 8 to 23 years in the female population and 11 to 22 years in the male population.



Digital ruler is extended to the point of measurement

Figure 4. Digital measurement acquisition. Digital measurements were performed on SSM-generated meshes. All meshes were registered with the same orientation, number of points, and 1-to-1 point matching, resulting in vertex IDs that represent homologous anatomical locations across meshes. The reference point (shown in green) corresponds to a fixed vertex index defining the trochlear groove and is consistent across all meshes. From these reference points, measurement lines were extended to points of interest, and distances were recorded in millimeters. IDs, identifiers; SSM, statistical shape modeling.

The mean age in the female MRIs was 15.8 (2.6) years, compared to 16.2 (2.3) years in the male MRIs. We used the SSM to generate representative distal femur reconstructions for each age between 13 and 19 years, enabling visualization of longitudinal changes in distal femoral morphology (Figure 5).

For physeal measurements (Figure 6), linear regression showed a significant positive association between age and each physeal distance measure in both sexes (all slope $P < .01$), indicating progressive physeal migration during adolescence. There were no statistically significant differences in physis location between the sexes at any age.

Linear regression analysis demonstrated a significant positive association between age and both medial and lateral trochlear depth in males and females (all slope p -values $< .01$), indicating progressive deepening throughout adolescence. At all ages, the lateral trochlea was significantly deeper than the medial trochlea. In addition, even after standardization for knee-size differences, we found that males exhibited a significantly deeper lateral trochlea than females after the age of 16 years (Figure 7).

Taken together, we observe that trochlear depth development is correlated with movement of the physis (Table 1). For both sexes, the correlation coefficient is greater than 0.85 for the medial, lateral, and anterior physis with both medial and lateral trochlear depth.

DISCUSSION

This study provides a detailed characterization of key structural changes in the distal femur that occur throughout adolescence. The application of a statistical shape model, combined with a 2-label approach to study the physis, enabled us to novelly model femoral shape and physis position independently of overall size. By removing global scaling effects through similarity registration, this approach allows identification of relative morphologic remodeling during adolescence rather than simple proportional enlargement of the distal femur with growth. Through correlation analysis and visual representations of age-specific femoral morphology, we observed that trochlear deepening and proximal physeal migration occur in parallel during skeletal maturation, reflecting coordinated changes in relative shape during adolescence.

This observation of proximal physeal movement, independent of increases in femoral scale, enhances our understanding of distal femoral growth and has potential clinical relevance. Awareness of physeal position at different developmental stages may aid in surgical planning by refining anatomical landmarks for trochleoplasty and related procedures.^{5,34} Because physeal location determines the region of safe epiphyseal bone available for surgical preparation, age-specific knowledge of its location may reduce the risk of iatrogenic growth disturbance.^{18,29} Furthermore, the parallel development of trochlear development and physeal migration provides a developmental 3D framework for understanding distal femoral maturation. These findings may inform future individualized modeling tools to assist with growth-aware surgical planning and improve care for skeletally immature patients.

The finding that the lateral depth of the trochlear groove is greater than the medial depth aligns with previous research on the patellofemoral joint analyzed via computed tomography and MRI.^{3,25,26} Our research builds on this existing literature by visually and quantifiably showing how this relationship is consistent among both sexes throughout adolescence. A normative 3D model of trochlear depth development, such as the one presented in this study, may serve as a future reference for identifying dysplastic trochleae and detecting patients at elevated risk of patellar dislocation at a younger age. The consistency between our results and previously reported anatomical features serves as a form of model validation, confirming that this framework accurately captures normative developmental anatomy. This represents a necessary first step before applying it to characterize abnormal trajectories, such as those seen in trochlear dysplasia, or to support future patient-specific surgical planning tools.

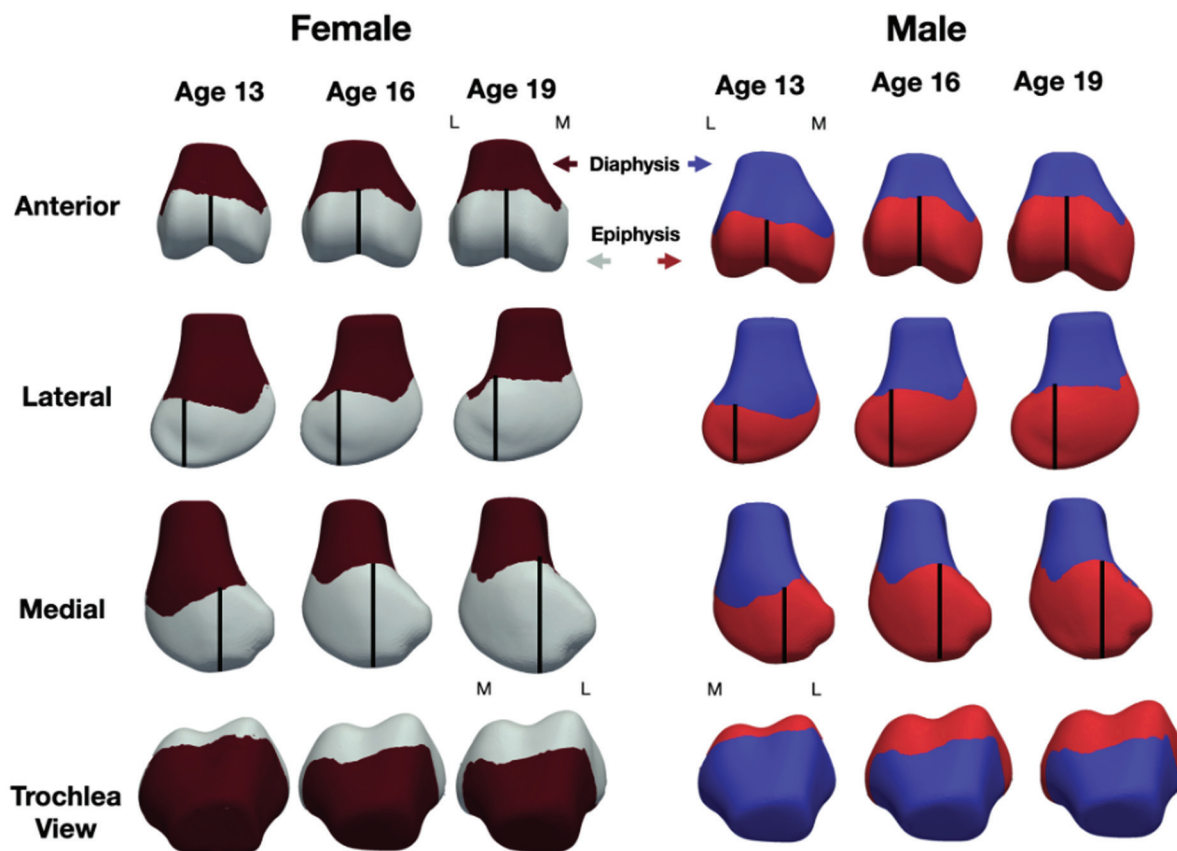


Figure 5. Average sex-based development of femur morphology. As shown by the anterior, lateral, and medial views, we observe proximal movement of the physis around the circumference of the femoral shaft. Additionally, the trochlear view demonstrates that as age progresses, the trochlear groove deepens, with the lateral side consistently positioned higher than the medial side. L, lateral; M, medial.

Previous research by Pagliuzzi et al²⁵ demonstrated that from 1 month to 10 years of age, males consistently exhibit a deeper trochlear groove than females. Ayala et al³ further showed that condylar and trochlear growth continues through age 11 and potentially beyond. However, it remained unclear whether this continued growth during adolescence would alter the observed sex-based differences in trochlear depth. Our findings extend previous knowledge by showing that, after age 16 years, the size-standardized male distal femur exhibits a statistically significant difference in the depth of the lateral trochlea compared with that of females. This difference in size-standardized depth after age 16 years aligns with the timing of female skeletal maturity, indicating that females typically reach maturity earlier and experience less subsequent growth than males.² Consistent with this concept, Tanaka et al³⁰ demonstrated that in adult MRI cohorts, reference values and diagnostic thresholds for trochlear morphology differ between skeletally mature males and females, underscoring the importance of sex-specific interpretation of trochlear measurements. While our measured differences in trochlear depth in the size-standardized femur may appear minor, it is essential to recognize that these measurements reflect only proportional changes. Both sexes show increases in

absolute femoral size with age, and males typically develop larger femurs throughout adolescence than females do.^{11,24} These absolute-size-based increases from age- and sex-related changes would further amplify our measured sex-based differences in trochlear depth.^{11,20,24}

We also examined the comparative development of the medial and lateral depths of the trochlea. Richmond et al²⁶ observed that medial and lateral trochlear depths develop at the same rate, resulting in no change in the medial-to-lateral ratio of trochlear depths over time in patients aged 2 months to 11 years. Our research extends the analysis into adolescence, demonstrating that in individuals with prototypical trochleae, both medial and lateral trochlear depths increase with age. These changes occur in parallel, with similar rates of deepening on both sides. These data suggest that the relative offset of lateral-to-medial trochlear depth observed in adulthood, a potential risk factor for patellar instability, can be identified as early as 13 years of age. These findings are supported by previous research conducted by Trivellas et al³¹ in a 2D MRI-based study. They similarly demonstrated that through adolescence, both medial and lateral trochlear height measurements increase linearly. These findings provide a developmental framework that may inform future investigations into growth-

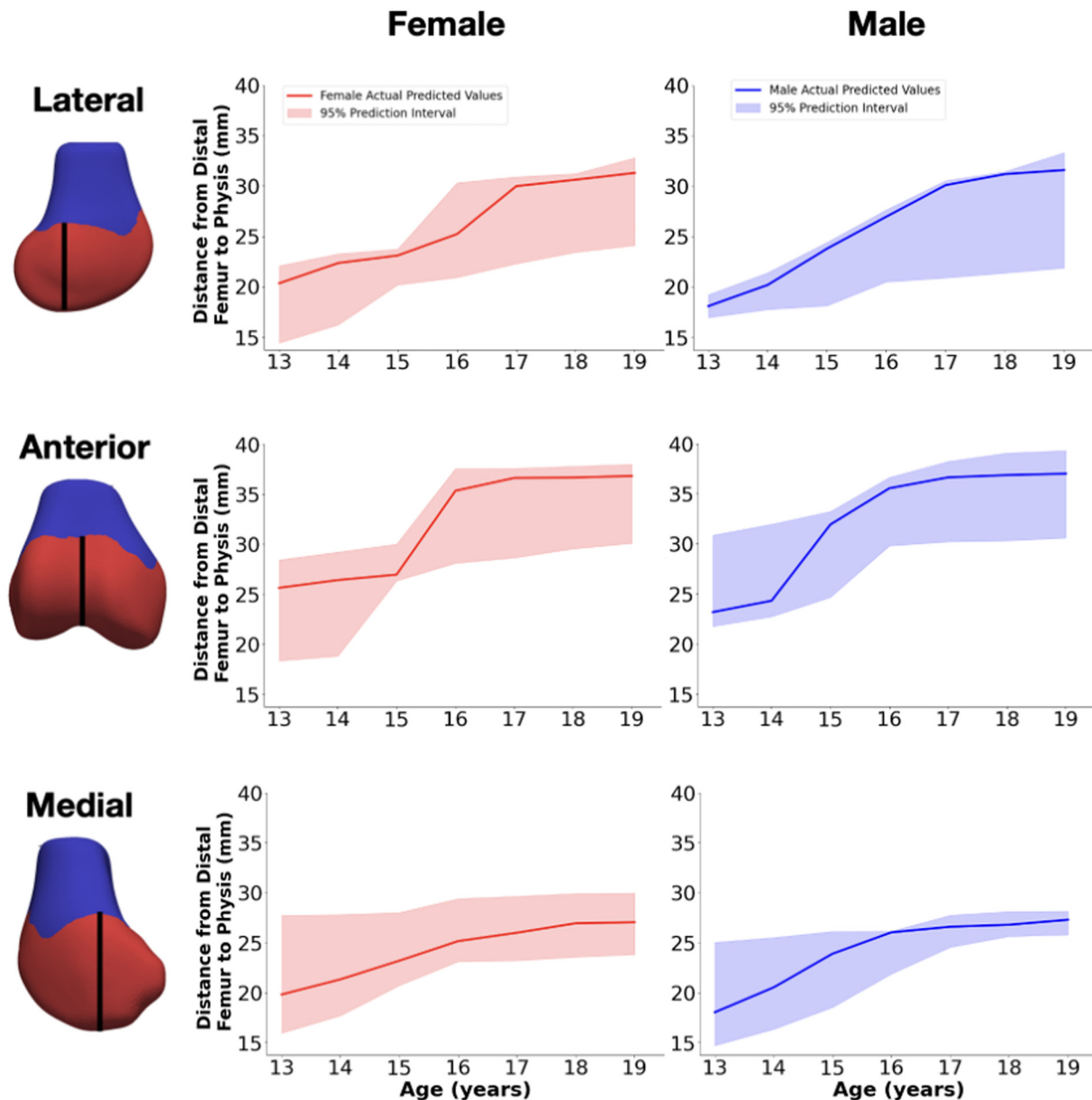


Figure 6. Measurements from the distal femur to the physis throughout development. Digital measurements from the size-standardized reconstructed femurs were used to quantify age- and sex-related changes in physis position. The solid line represents the regression-predicted mean value at each age. Specifically, for each age, we applied the mean regression coefficients from the sex-specific linear model to the SSM to generate a single mean prototypical 3D mesh that reflects the mean shape trajectory across the dataset. Measurements extracted from these mean meshes are shown as the solid line. The shaded region reflects the 95% confidence band generated by varying the shape score coefficients within their standard errors and reconstructing the corresponding meshes. Because measurements are extracted from the reconstructed meshes, the confidence band is not symmetric around the solid line. SSM, statistical shape modeling; 3D, 3-dimensional.

modulating strategies in younger skeletally immature populations, given that the medial-to-lateral relationship appears to remain stable across development in our cohort of anatomically normal patients.

Guided growth techniques, which stimulate or restrict growth in targeted regions of the physis and periarticular structures, have already shown success in skeletally immature hip dysplasia, leading to reduced instability, more

normal joint development, and lower rates of osteoarthritis compared with no intervention.^{9,32} More recently, advances in interventional radiology have enabled precise, localized modulation of physeal growth through thermal-based methods, such as microwave ablation.^{6,23} These minimally invasive strategies present a potential pathway to proactively correct dysplastic development while preserving growth. The strong correlations between physeal migration and

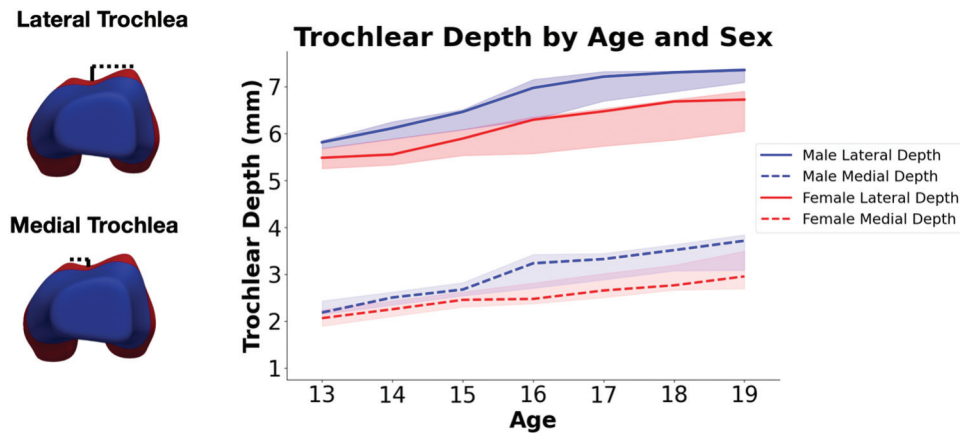


Figure 7. Measurements from trochlear depths throughout development. Values represent regression-predicted mean trochlear depth derived from sex-specific linear models relating age to trochlear morphology. The solid lines depict the depth in the model-generated mean shapes at each age, and the shaded regions represent the corresponding 95% confidence bands of the regression model. Measurements were obtained from SSM-generated meshes after removing the overall scale via similarity registration. SSM, statistical shape modeling.

TABLE 1
Correlation Coefficients Between Trochlear Growth and Physis Movement^a

Sex	Region	<i>r</i> Lateral Physis	<i>r</i> Anterior Physis	<i>r</i> Medial Physis
Male	Lateral trochlea	0.99 ^c	0.97 ^c	0.98 ^c
	Medial trochlea	0.98 ^c	0.94 ^c	0.95 ^c
Female	Lateral trochlea	0.96 ^c	0.96 ^c	0.99 ^c
	Medial trochlea	0.95 ^c	0.86 ^b	0.95 ^c

^aAll relationships demonstrate strong or very strong correlation, highlighting that, during development, physis location is strongly associated with the deepening of the trochlear groove.

^b $P < .05$.

^c $P < .001$.

trochlear depth observed in this study highlight the need for future investigations to determine whether targeted modulation of specific physeal regions could causally influence trochlear concavity. Such work may represent a potential avenue for growth-preserving interventions in the management of trochlear dysplasia.

This study has limitations. As a proof-of-concept investigation, it was designed to highlight SSM's potential as a method for depicting normative developmental trajectories. While MRI data were initially requested for patients aged 8 to 23 years (a 15-year range), the 10th to 90th percentiles of our data spanned ages 13 to 19 years. To prevent overinterpretation of results in age groups with minimal data, we analyzed only data from ages 13 to 19 years. However, this age group reflects mid-to-late adolescence, during which remaining growth potential, particularly in females, may be limited. Future studies incorporating younger age groups are needed to characterize better early developmental changes and their potential relevance to surgical procedures. Another limitation relates to the linear regression used to model growth. While this approach provides interpretable trajectories,

skeletal development is not strictly linear, and this may introduce bias. Additionally, while knees with trochlear dysplasia were excluded, it is important to acknowledge that the knees in this study were derived from clinical MRIs. Abnormal function in other regions of the knee could potentially influence trochlear growth patterns. Future studies could confirm these findings by using a research population of patients with no preexisting knee joint abnormalities. Furthermore, MRI acquisition parameters, including magnetic field strength and sequence parameters, varied across the 14-year collection window and among clinical settings, potentially affecting segmentation consistency and downstream shape modeling results. This study focused on mapping normal trochlear growth, representing a necessary first step toward understanding dysplastic development. We acknowledge that, as a cross-sectional study, the age-related differences observed here are assumed to represent developmental change over time. Still, longitudinal confirmation of these trajectories has not been performed. Growth patterns in dysplastic trochleae, which may follow distinct trajectories, have not been investigated and remain an important direction for

future work. Further, our segmentations focused exclusively on osseous shape and morphology. Cartilage was not segmented; therefore, our model does not evaluate cartilaginous morphology. Finally, our methodology assessed physeal positioning only at the femoral surface. Internal physeal morphology may exhibit different relationships with trochlear development, but given the surface-based mesh extraction employed here, these features could not be captured.

The present study provides standardized developmental trajectories using SSM, offering a reference framework against which dysplastic morphology can be compared in future investigations. Future work should expand this framework to include younger populations, where growth-related surgical considerations may be more clinically relevant. Extending this approach to comparative analyses of dysplastic and anatomically normal populations may help identify timepoints at which developmental trajectories diverge. In turn, this may flag individuals whose morphology has deviated sufficiently to warrant early growth modulating intervention. Beyond candidate identification, the generative nature of the SSM could help localize the specific regions and patterns of abnormality in a patient's trochlear morphology, informing where and how to intervene. In a longitudinal setting, this framework could also serve as a foundation for patient-specific monitoring tools, allowing surgeons to track morphologic response to treatment and assess whether a patient is returning toward a normative developmental trajectory.

CONCLUSION

This study used SSM to characterize prototypical trochlear development across adolescence, enabling sex-specific analysis of morphological differences. Our results suggest that trochlear deepening and proximal pysis movement occur in parallel throughout development, and that sex-based differences in trochlear depth persist into adolescence. These findings build upon observations in younger populations. They may inform surgical decision-making in procedures such as trochleoplasty, while also supporting the development of early identification and growth-modulating interventions, such as guided growth, for patients at risk of patellar instability.

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APPENDIX

MRI DESCRIPTION

Magnetic resonance imaging (MRI) scans were acquired across multiple sites and time points from 2009 to 2023, with 35 scans performed on a 1.5T scanner and 226 on a 3T scanner. Where available, dedicated knee coils were used; however, as scans were obtained from a variety of

clinical settings, coil type was not uniformly recorded across all studies. Despite this variation, all scans maintained sufficient image quality for consistent segmentation, as reflected by the narrow range of in-plane resolution across the cohort. Descriptive statistics of the acquisition parameters of the MRIs used in this study are outlined in Table A1.

AUTOMATED SEGMENTATION VIA NNUNET

To aid in the iterative semi-automated approach to segmenting each MRI, we trained an nnUNet-based deep learning model³ to segment the diaphysis and epiphysis. Initially, we trained our automated segmentation model on 25 manually segmented MRIs. Subsequently, the model was applied to 100 unlabeled MRIs, which were then manually reviewed and corrected. With this enhanced dataset of 100 MRIs, we retrained the model and applied it to segment the entire dataset. Each segmentation was manually inspected, and corrections were applied as necessary.

To mitigate overfitting, model training was stopped once the loss plateaued (200 epochs). Our optimal model achieved an overall Dice similarity coefficient of 0.96 across both the epiphysis and diaphysis labels on a test dataset of 20 manually segmented MRIs (1 represents perfect agreement between machine-generated and manual segmentations).

STATISTICAL SHAPE MODEL

Statistical Shape Model Fitting

Our shape modeling pipeline used previously established methods for surface meshing, registration, and feature extraction. To more efficiently use all data, all left knees were flipped along the medial-lateral axis to obtain right knees for shape modeling.

Surface Meshes. Our final representation used for fitting the shape model was a 3-dimensional (3D) surface of the distal femur, with surface nodes labeled as belonging to the diaphysis (and physis) or the epiphysis. To generate this representation, we first combined the diaphysis and epiphysis labels into a single binary label mask. To smooth bone surfaces and reduce slicing artifacts, we applied a Gaussian filter with a variance of 5 to the binary masks and then extracted surfaces using the marching cubes algorithm.⁴

Registration. To learn shape features, we found matching anatomical points between all patients in the dataset. First, to remove the effect of size, all surfaces were similarity registered (rigid + isotropic scale) to a reference knee, the mean knee in the dataset. Then, a nonrigid registration routine using spectral coordinate-based features was used to find matching points between each surface.^{1,2,5}

Shape features. The mean matching points across all patients were calculated and subtracted from each individual's shape, yielding individual deviations from the mean shape. Principal components (shape features) were learned by applying principal component analysis to these data.

TABLE A1
MRI Acquisition Parameters and Spatial Resolution of Included Scans^a

Measurement	Mean	SD	Minimum Value	Maximum Value
Flip angle, deg	128.72	19.06	75	160
Echo time, ms	33.4	8.10	6.58	41.19
Slice thickness, mm	2.55	0.40	2.00	4.00
Repetition time, ms	3904.69	828.95	886	7555.31
X-dimension in plane resolution, mm	0.2925	0.0377	0.1984	0.625
Y-dimension in plane resolution, mm	0.2925	0.0377	0.1984	0.625

^aWhile MRI protocols varied across patients and timepoints, all sequences fell within a narrow range that maintained sufficient spatial resolution and image quality for consistent analysis. The in-plane resolution was highly consistent across scans, supporting accurate segmentation and statistical shape modeling. MRI, magnetic resonance imaging.

Segmentation quality control. Before creating our final statistical shape modeling (SSM), we generated preliminary SSMs to identify and screen potential segmentation failures. Outliers were identified based on each mesh's principal component scores and the L_1 norm of all principal components. An outlier was defined as any mesh with an individual principal component or an L_1 norm that deviated from the mean by >3 standard deviations. All outliers were manually reviewed, segmentations were corrected where appropriate, and participants were excluded if their sulcus angle was $>180^\circ$.

Finding Statistical Shape Model Settings

Our purpose was to model prototypical aging in male and female adolescents. However, it was unclear if variations in sex were distinct enough to require sex-specific shape models. To determine whether sex-specific shape models outperformed a single shape model on mixed-sex data, we conducted an ablation study by training female-only, male-only, and mixed-sex shape models on progressively larger datasets. We then applied this model to reconstruct the femurs of held-out female ($n = 20$) and male ($n = 20$) patients.

Applying the shape model to the test set included optimizing the principal components of the trained shape model to best reconstruct a given example. First, the example was similar to the SSM's mean shape. Then, the SSM principal components were optimized to minimize a Chamfer loss between the original and reconstructed meshes using the Adam optimizer in PyTorch. Final surface reconstructions were evaluated by computing the average symmetric surface distance (ASSD) between the original and SSM-reconstructed surfaces.

We evaluated the ASSD of reconstructed meshes using either a mixed-sex SSM or sex-specific SSMs to determine which model could most accurately depict prototypical relations (Figure A1). Our results showed that for test sets comprising male or female meshes, the mixed-sex SSM achieved the lowest (best) ASSD compared with the sex-specific models (Figure A1). This finding suggests that sex-based differences were not a significant factor in the surface reconstruction of meshes. Furthermore, as the number of meshes used to train the SSM increased,

reconstruction accuracy generally improved, thereby reducing ASSD. Based on these results, we determined that a mixed-sex SSM that uses all available data provides optimal surface reconstruction for patient meshes.

3D MESHES OF THE DISTAL FEMUR

Prediction of Held Out Meshes via Lasso Regression

To validate our findings, we conducted additional tests using the same SSM. First, we assessed whether increasing the number of knees in the regression model would improve its performance in predicting patient age. We also evaluated whether sex-specific regression models or a mixed-sex regression model would more accurately predict patient age. To prevent overfitting, the regression model was regularized with Lasso.

Our results showed that the regression model more accurately predicted the age of held-out data when trained within a specific sex category than when trained in mixed-sex or opposite-sex settings (Figure A2). This finding suggests distinct and unique developmental timelines between male and female knees. Additionally, consistent with the SSM training, including more patient data in the regression model's training set generally improved its performance.

Obtaining Measurements Using Generated Age-Specific Prototypical Knees

To help standardize measurements, unique point indices were identified on each mesh to ensure the same starting point was used every time. Since the formation of the SSM aligned each femur along a common coordinate system, across meshes representing different ages, the same point index always corresponds to the same area of interest, such as the highest point of the lateral trochlea. Measurement would begin from the corresponding point index and end at the region of interest, using the built-in ruler from Paraview. All measurements were obtained from size-normalized knees. For scale reference of all generated knees, the medial-lateral width of the size-normalized femoral condyle measured 72.85 mm.

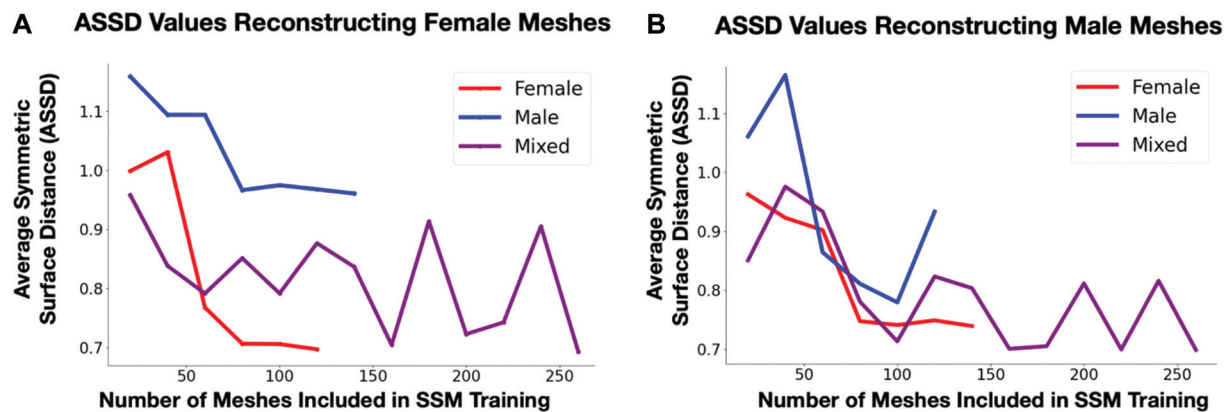


Figure A1. Testing mesh reconstruction. (A) ASSD values for reconstructing female meshes using SSMs trained on female-only, male-only, and mixed datasets. The mixed and female SSMs generally yielded lower ASSD values, indicating more accurate reconstructions of female morphology. (B) ASSD values for reconstructing male meshes using the same three SSM types. As with the female meshes, the mixed SSMs consistently produced the lowest ASSD values, suggesting they better capture anatomical variation across both sexes. ASSD, average symmetric surface distance; SSM, statistical shape modeling. Lower ASSD values correspond to more accurate reconstructions and indicate that the SSM more effectively captures the underlying shape variability. Therefore, we elected to use mixed model SSMs, as they yielded the lowest ASSD across both sexes.

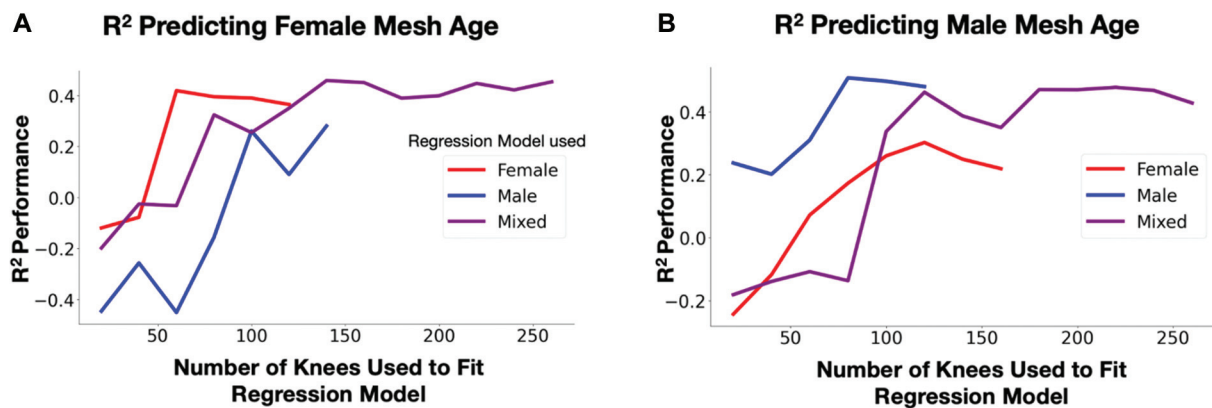


Figure A2. Testing mesh age prediction. (A) R^2 values for predicting age in female meshes using regression models trained on female-only, male-only, or mixed-sex datasets. (B) R^2 values for predicting age in male meshes using the same 3 regression model types. In all tests, a mixed-sex SSM was used to isolate the effect of training data composition on regression performance. As the number of meshes used in regression training increases, R^2 generally improves, demonstrating that larger training sets provide greater predictive power. Additionally, models trained on meshes of the same sex as the test set yield higher R^2 values than those trained on meshes of the opposite sex, suggesting distinct developmental patterns between male and female adolescent knees. SSM, statistical shape modeling.

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